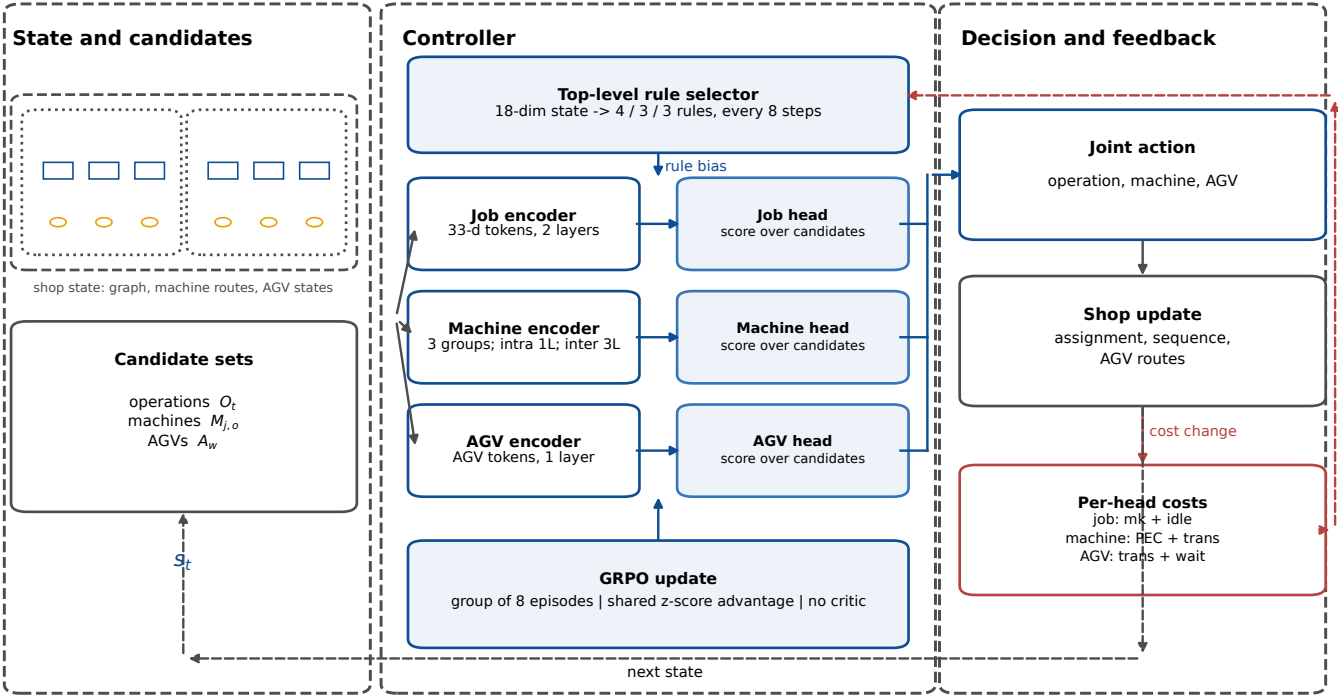

State and candidates
shop state: graph, machine routes, AGV states
Candidate sets
operations O_t
machines $M_{j,o}$
AGVs A_w
s_t
Controller
Top-level rule selector
18-dim state -> 4 / 3 / 3 rules, every 8 steps
rule bias
Job encoder
33-d tokens, 2 layers
Job head
score over candidates
Machine encoder
3 groups; intra 1L; inter 3L
Machine head
score over candidates
AGV encoder
AGV tokens, 1 layer
AGV head
score over candidates
GRPO update
group of 8 episodes | shared z-score advantage | no critic
Decision and feedback
Joint action
operation, machine, AGV
Shop update
assignment, sequence,
AGV routes
cost change
Per-head costs
job: mk + idle
machine: REC + trans
AGV: trans + wait
next state
encoder
head
cost
state / shop